

Movies Data Analysis Report

Author : ['Sufiya Ali']

Date: 8-june-2026

Tool used: Python, Pandas, Numpy, Matplotlib & Seaborn

1.Objective:

The objective of this project is to analyze a dataset of 2,000 movies to uncover patterns in ratings, languages, and audience engagement. The goal is to identify top-performing and underperforming movies using a custom Engagement Score.

2.Dataset Overview:

Total records	2,000 Movies
Total columns	8
Data range	1938 to 2028

Column	Description
Id	Unique movie ID
title	Movie name
original_language	Language of movie
overview	Movie description
release_date	Release date
original_title	Title in original language
vote_average	Rating out of 10
vote_count	Total votes received

3.Data Cleaning:

- 4 missing values in 'release_date' column → ignored
- vote_count = 0 movies → removed using filter (vote_count > 500)

- Unreliable movies removed to ensure trustworthy results
- 15 missing values in 'overview' column → ignored

4.Key Insights:

Insight 1 — Ratings Distribution:

- Average Rating: 6.67 out of 10
- Most movies fall between 6 to 8 rating range

Insight 2 — Language Breakdown:

- **English** : 73.5% (1470 movies)
- **Japanese** : 4% (80 movies)
- **Korean** : 3.4% (69 movies)
- **Chinese** : 2.5% (51 movies)
- **French** : 2.3% (46 movies)

Insight 3 — Engagement Score:

- Formula: $\text{vote_average} \times \log_{10}(\text{vote_count})$ Combines both quality and popularity for fair ranking

Insight 4 — Top 5 Movies:

1.	The Shawshank Redemption	39.10
2.	Interstellar	38.99
3.	The Dark Knight	38.85
4.	Inception	38.46
5.	Fight Club	38.02

Insight 5 --- Worst 5 Movies:

1.	Human Centipede 3	11.06
2.	War of the Worlds	12.31
3.	Snow White	13.87
4.	Hellboy Crooked Man	14.25
5.	Mortal Kombat Ann.	14.26

5. Conclusion:

- This project analyzed 2,000 movies using Python and Pandas. A custom Engagement Score ($\text{vote_average} \times \log_{10}(\text{vote_count})$) was built to fairly rank movies by combining quality and popularity. Results show that classic movies like Shawshank Redemption dominate due to high ratings and massive votes.